

Table of Contents

Cisco Secure Workload — NIST SP 800-171 Rev. 3

Reference Design | CUI in Nonfederal Systems

Version: 1.0

Standard: NIST SP 800-171 Rev. 3 (May 2024), *Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations*

Use Case: Fresh install, hybrid environment (on-premises + cloud), CUI processing enclaves

Reader's Guide

Who this is for. Nonfederal organizations that process, store, or transmit CUI under federal contracts or agreements; security architects building CUI enclaves; and assessors collecting technical evidence for 800-171A assessment procedures.

Questions this reference design helps you answer:

- *For requirement 03.13.06 (deny by default, allow by exception), can I show continuous enforcement at the workload tier—not only perimeter firewalls—with a documented allow-list derived from observed traffic?*
- *For 03.01.03 (information flow enforcement for CUI), can I prove authorized flows between CUI and non-CUI segments and detect or block unauthorized cross-boundary paths?*
- *For 03.03.x (audit and accountability), can I export flow and process-level telemetry that traces CUI-adjacent workloads to specific processes and time windows for SIEM retention?*
- *For 03.04.x (configuration management), can I baseline communications and software exposure, then demonstrate drift when new services, ports, or binaries appear on CUI systems?*
- *For 03.11.x (risk assessment) and vulnerability prioritization, can I combine CVE exposure with east-west reachability and EPSS-style context to support POA&M and risk narratives?*

What you'll need. A current CUI boundary definition (which components process, store, or transmit CUI), a system security plan (SSP) or equivalent scope description, labeling standards for workloads, and assessor or PMO expectations for evidence format (reports, exports, screenshots).

Where to start. Sections 1–2 for scope alignment; 3–6 for deployment and baseline; 7–9 for policy design tied to 03.01.03, 03.01.05, and 03.13.x; section 13 if you are packaging evidence for an assessment window.

CSW primer — if you are new to Cisco Secure Workload

Cisco Secure Workload (CSW) is a **workload protection platform**. A lightweight **agent** on each server/VM/container observes processes and network flows; **cloud connectors** add AWS/Azure/GCP inventory where agents are not deployed.

CSW term	Meaning	Why compliance teams care
Scope	Logical boundary (CDE, PHI zone, CUI enclave)	Defines the systems you must prove are isolated
Label / filter	Tag or query assigning workloads to scopes	Automates scope membership — reduces drift
ADM	Application Dependency Mapping from observed traffic	Live network/data-flow diagram for assessors
Workspace	Policy container for one scope or application	Where allow/deny rules are authored
Monitor → Simulation → Enforce	Safe rollout sequence	Simulation = change-board evidence before blocking traffic
Denied Connections	Log of flows blocked by policy	Primary proof that enforcement operates

Console areas: Investigate (inventory, flows, vulns) · Defend/Segmentation (policy) · Manage (agents) · Platform (connectors) · Administration (audit log)

Read next: [Compliance evidence playbook](#) (full step-by-step) · [About CSW](#) (platform intro)

Universal evidence workflow

Execute these phases for **this framework's** compliance boundary. Map exports to control IDs in the **Reporting & Evidence** section below.

Phase	Goal	Key CSW actions	Typical duration
1 — Coverage	Every in-scope workload in CSW	Install agents/connectors ; apply compliance	Days 1–10

Phase	Goal	Key CSW actions	Typical duration
		labels; create scope	
2 — Baseline	Machine-generated flow map	Run ADM ≥ 2 weeks; export clusters; app-owner signoff	Days 11–28
3 — Policy	Designed isolation before enforce	Build workspace; Simulation mode; fix false positives	Days 29–45
4 — Operate	Continuous proof between audits	Enforce; quarterly export pack; ADM refresh every 90 days	Ongoing

Phase 1 checklist (coverage)

- ☐ In-scope host list reconciled to CSW Inventory (100% or documented exceptions)
- ☐ Labels applied: `compliance:<framework>`, `data:<classification>`, `env:<tier>`
- ☐ CSW scope created matching compliance boundary
- ☐ **Export:** inventory CSV + agent status screenshot

Phase 2 checklist (baseline)

- ☐ ADM running on compliance scope for full business cycle (≥ 2 weeks)
- ☐ Unexpected flows documented (shadow IT, vendor egress, scope creep)
- ☐ App owners signed cluster-to-application mapping
- ☐ **Export:** ADM diagram + flow samples with process context

Phase 3 checklist (policy)

- ☐ Default-deny posture defined for sensitive scope
- ☐ ADM-imported rules refined; Simulation run ≥ 1 week
- ☐ Change tickets for false-positive fixes; exception register updated
- ☐ **Export:** policy export + simulation report

Phase 4 checklist (operate)

- ☐ Enforcement enabled on pilot scope; negative test recorded in Denied Connections
- ☐ Quarterly pack: inventory, policy, denies, vulns, audit log (see playbook)
- ☐ SIEM integration verified (sample events)
- ☐ **Export:** enforcement screenshot + quarterly binder

What CSW evidence does not replace

Physical access, HR/training records, encryption key management, signed BAAs/vendor contracts, formal pen tests, and assessor attestation still require separate programmes. CSW

addresses the **workload-resident** slice: segmentation, flows, process context, vuln reachability, and change drift.

CSW effectiveness for this framework

- CUI enclave isolation with 03.01/03.13 flow enforcement
- CMMC L2 underpinning evidence
- 800-53 Rev 3 family alignment for assessors

Compared to manual programmes: static diagrams and annual firewall samples age immediately; CSW ties evidence to **live workload behaviour** and produces queryable exports on demand — supporting "operating effectively" language in PCI v4.0, SOC 2 CC7, and HIPAA risk analysis.

1. Overview

NIST SP 800-171 Rev. 3 modernizes the CUI protection requirements, aligns requirement numbering and structure with **NIST SP 800-53 Rev. 5**, and preserves the core outcome: confidentiality of CUI in nonfederal systems. Cisco Secure Workload (CSW) contributes **technical and operational evidence** for requirements where workload visibility, micro-segmentation, vulnerability context, and forensic telemetry are relevant. It does not replace administrative, physical, or personnel controls, identity providers, or cryptographic implementations.

1.1 Alignment with NIST SP 800-53 Rev. 5 and CMMC

Rev. 3 requirements include source-control references to 800-53 (for example, 03.13.06 maps from SC-7(5)). For a detailed control-family treatment mapped to CSW at the 800-53 control level, see the companion reference design below. **CMMC 2.0** uses NIST SP 800-171 as its technical basis for Level 2; contractors should **confirm which revision (Rev. 2 vs. Rev. 3)** their contracts, PMO, or assessment framework specify during any transition period. This reference design is written to **Rev. 3** identifiers (e.g., 03.01.03).

1.2 Rev. 3 Family Structure and CSW Relevance

Family (Rev. 3)	Title	CSW relevance
03.01	Access Control	Direct — identity-aware and scope-based micro-segmentation; deny-by-default enforcement paths
03.02	Awareness and Training	Out of scope — HR/training systems
03.03	Audit and Accountability	Direct — flow + process telemetry; export to

Family (Rev. 3)	Title	CSW relevance
		SIEM
03.04	Configuration Management	Direct — software inventory, ADM baselines, drift versus approved communication profiles
03.05	Identification and Authentication	Supporting — enforcement of authenticated paths (e.g., Kerberos/LDAPS-only); IdP configuration remains outside CSW
03.06	Incident Response	Supporting (evidence) — forensics, timelines, scope of compromise views
03.07	Maintenance	Supporting — monitoring of maintenance-related connectivity if expressed in policy baselines
03.08	Media Protection	Limited — no physical media control; logical exfiltration paths may be constrained by policy
03.09	Personnel Security	Out of scope
03.10	Physical Protection	Out of scope
03.11	Risk Assessment	Direct — CVE visibility, reachability from threat-facing workloads, prioritization inputs
03.12	Security Assessment and Monitoring	Supporting — continuous monitoring artifacts; CSW does not replace formal control testing
03.13	System and Communications Protection	Direct — CUI enclave isolation, boundary enforcement, plaintext/unauthorized protocol detection

Family (Rev. 3)	Title	CSW relevance
03.14	System and Information Integrity	Direct — process monitoring, anomaly signals, forensic integrity of telemetry
03.15	Planning	Supporting — inventory and scope evidence feed SSP accuracy
03.16	System and Services Acquisition	Limited — monitor connectivity to external services; contract terms remain organizational
03.17	Supply Chain Risk Management	Limited — workload component visibility; full SCRM is organizational

2. Pre-Deployment Checklist

Before deploying CSW sensors into CUI or CUI-adjacent enclaves, confirm the following:

- ☐ CSW cluster (SaaS or customer-managed) is provisioned; data residency and IRP for telemetry meet organizational and contractual rules for CUI.
- ☐ Outbound connectivity from workloads to the CSW cluster (typically TCP 443) is allowed and documented in the SSP connection table.
- ☐ Linux and Windows sensor versions are approved for all target OS builds in the CUI baseline.
- ☐ Cloud connectors (AWS, Azure, GCP) are authorized if agentless visibility is used; IAM roles / service principals are least-privilege and documented.
- ☐ Stakeholders engaged: ISSO, system owner, network operations, contracting/CUI program office.
- ☐ Change windows approved; **initial mode is monitoring**, not enforcement, until ADM and policy simulation complete.

3. Phase 1 — Sensor Deployment (Days 1–5)

3.1 On-Premises Workloads

Install software sensors:

```
# Linux (RHEL / CentOS stream / Ubuntu variants – use package matching
your distro)
sudo rpm -ivh tet-sensor-<version>.rpm      # RHEL family
# or
```

```
sudo dpkg -i tet-sensor-<version>.deb      # Debian family

# Verify agent service (name may vary slightly by release; consult
# release notes)
sudo systemctl status tet-engine || sudo systemctl status csw-agent
```

Configuration guardrails for CUI:

- Enforcement mode: **Monitoring only** until policy sign-off.
- Enable process hash, connection telemetry, and vulnerability exposure features supported on your agents.
- Apply labels at install time: cui_scope, enclave, environment, owner.

3.2 Cloud Workloads

Option A — Agent on every CUI-bearing instance: same as on-prem.

Option B — Cloud connector (complementary):

CSW UI → Platform → External Orchestrators
→ Add AWS / Azure / GCP connector
→ Attach read-only discovery role (EC2/VM inventory, optional flow log paths per vendor)
→ Map cloud tags → CSW labels (e.g., tag CuiEnclave=ALPHA → scope CUI-ALPHA)

3.3 Validation

CSW UI → Manage → Agents
→ Status: Active for every in-scope workload
→ Version alignment with approved standard image
→ Confirm flows visible within 15–30 minutes of agent start

4. Phase 2 — Scope Design and CUI Inventory (Days 6–12)

4.1 Recommended Scope Hierarchy

Model **CUI enclaves** explicitly; never rely on a single flat “production” scope for assessment evidence.

```
Root
├── Organization
│   ├── CUI-Enclaves
│   │   ├── CUI-Tier1-Apps
│   │   ├── CUI-Datastores
│   │   └── CUI-Integration-Brokers
│   └── Security-Protection-Assets
│       ├── Identity (read-only policy exceptions)
│       ├── Patch-Repo / Artifactories (as approved)
│       └── SOC-SIEM-Collectors
```

- └─ Backup / DR targets
- └─ Non-CUI (or “Corporate”)
 - └─ Dev/Test (must not share policy workspace with CUI if contract requires segregation)
 - └─ General IT

4.2 Label Schema (Minimum)

Label key	Example values	Purpose
cui_scope	in_scope, spa, out_of_scope	Aligns workload to SSP boundary
cui_role	processes, stores, transmits	Maps to CUI handling roles
enclave_id	ENCLAVE-A	Groups policies and ADM workspaces
data_class	cui, internal, public	Drives segmentation rules
owner_team	prog-office-xyz	Accountability in exports

4.3 Discovery Filters (Examples)

Filter: CUI-Candidate-DBs

- Process: postgres, oracle, mysqld, sqlservr, mongod
- Ports: 5432, 1521, 1433, 3306, 27017

Filter: CUI-Candidate-App-Tiers

- Tag from cloud: DataClassification=CUI
- Hostname pattern: *cui*, *gov*, *fed*

Filter: Common-Exfil-Protocols (review only – policy may block after baseline)

- Outbound destination not in approved list AND process in (curl, wget, ftp, scp)

5. Phase 3 — Visibility and Baseline (Days 10–25)

5.1 Application Dependency Mapping (ADM)

ADM establishes the **authorized communication baseline** needed for 03.01.03, 03.04.x comparison, and 03.13.06 allow-lists.

CSW UI → Investigate → Application Dependency Mapping

- New Workspace
- Name: CUI-ADM-<Enclave>-<YYYYQQ>
- Scope: CUI-Enclaves (or single enclave for stricter separation)
- Duration: ≥ 14 days (include month-end / payroll / batch cycles if

applicable)

→ Enable rich process context and user context where supported

During review, document:

Question	Relevance
Which workloads initiate sessions from non-CUI into CUI?	03.01.03, 03.13.01
Which CUI workloads initiate to non-CUI (break-glass, SaaS, email)?	03.01.03, 03.01.20
Where is traffic cleartext (HTTP, LDAP, FTP) versus encrypted?	03.13.x, 03.13.08 (as applicable)
What is the blast radius if a given internet-facing tier is compromised?	03.11.x

5.2 Software Inventory and Drift

CSW UI → Investigate → Inventory / Vulnerability (per product navigation)

→ Scope: CUI-Enclaves

→ Export: workload, package/CVE summary, listening ports, last seen

Re-run ADM **at least quarterly** or after major releases; treat newly observed flows as **configuration changes** under 03.04.x.

6. Phase 4 — Policy Design (Days 22–40)

6.1 Policy Workspace Pattern

CSW UI → Defend → Segmentation

→ New Workspace: CUI-<Enclave>-Enforcement

→ Scope: matching CUI scope from section 4

→ Import ADM-suggested rules → translate to explicit allow/deny with change tickets

→ Mode: Simulation → Enforcement (staged)

6.2 Illustrative Policy Stanzas (Conceptual)

Translate organizational policy into CSW rules. Examples below are **patterns**; port lists and peers must come from your ADM and SSP.

03.13.06 — Deny by default, allow by exception (east-west at workload tier):

Implicit default: DENY unmatched intra-scope traffic after cutover

Explicit allows (examples):

ALLOW: CUI-App-Tier → CUI-DB-Tier [TCP 443, 5432 from approved SG/workload set only]

ALLOW: CUI-App-Tier → Identity [TCP 636, 88, 464 – Kerberos/LDAPS paths per your design]

ALLOW: SOC-Collector → All-CUI [TCP 443 agent telemetry – if architected this way]
DENY: Non-CUI → CUI [except documented jump hosts / service accounts]
DENY: CUI → Internet [except explicit proxy or vetted SaaS egress IPs]
LOG: Any → CUI unmatched traffic (alert + SOC review)

03.01.03 — Information flow enforcement between CUI and non-CUI:

Separate policy chain or rule section titled "CUI-Boundary-Flow"
ALLOW: Approved-Integration-Host ↔ Partner-VDI [documented in connection agreement]
DENY: CUI-Datastores → Non-CUI-Desktops (default)
ALLOW: CUI-Datastores → Backup-Appliance [narrow destination list]

03.01.05 — Least privilege (network dimension):

- Every ALLOW rule has: business owner, ticket ID, expected process set, expiry/review date.
- Prefer **workload identity** (agent visibility) over coarse subnet-only rules.

6.3 Plaintext and Unexpected Protocol Handling (Support to 03.13.x)

Create alerts for flows that indicate cleartext or risky transports from CUI scopes (examples: HTTP to sensitive ports, LDAP:389 from CUI app tiers, unapproved RDP paths). Pair technical blocks with compensating monitoring where applications cannot yet migrate to TLS.

7. Phase 5 — Enforcement and Validation (Days 40+)

Stage	Mode	Objective
1	Simulation (2+ weeks)	Tune alerts; no production impact
2	Selective enforcement	Enforce obvious deny rules (e.g., CUI → open internet)
3	Full enforcement	Default deny with signed allow-list; break-glass process documented

Break-glass: maintain an auditable process (ticket + time-bound policy toggle or scoped exception) for emergency change; avoid permanent “ANY ANY” exceptions.

8. Phase 6 — Risk Assessment and Vulnerability Context (Ongoing)

CSW UI → Investigate → Vulnerability Report

- Scope: CUI-Enclaves
- Sort by environmental priority (e.g., critical CVE on internet-adjacent + ADM path to datastore)
- Export for POA&M / risk register

Use **reachability** from ADM plus exposure to justify remediation order (supports 03.11.x narrative). Where patching is delayed, document **compensating** CSW controls (tightened source IP, removal of path, extra logging on affected tier).

9. Phase 7 — Audit Readiness: Monitoring and Accountability (Ongoing)

9.1 Alerts Suited to CUI Operations

Alert theme	Example trigger	Typical requirement tie-in
Boundary violation	Non-approved source connects to CUI tier	03.01.02, 03.13.01
Flow policy deny hit	Enforcement drop / reject	03.13.06
Novel lateral path	New east-west service on CUI datastore	03.01.03, 03.04.x
Cleartext indicator	HTTP/LDAP from CUI app to sensitive listener	03.13.x
Sensor gap	Agent offline on CUI server	03.03.x (visibility gap)

9.2 Forensic Export Pattern

CSW UI → Investigate → Flow Search

- Time UTC window, source/destination workload, action
- Export CSV/JSON per SOC reference design

CSW UI → Investigate → Process Search

- Process tree for suspect host during same window
- Attach exports to incident record

Integrate alerts and enriched logs with SIEM (syslog, API, or vendor connector per your subscription).

10. Control-by-Control Mapping (Representative Set)

The table below links **selected** Rev. 3 requirements to **CSW capabilities** and **example evidence**. It is not exhaustive; complete coverage requires mapping every applicable requirement in your SSP and 800-171A assessment procedures.

Requirement	Title / intent (summary)	CSW capability	Evidence produced
03.01.02	Access enforcement	Scope-based micro-segmentation; deny default with explicit ALLOW	Policy workspace export; enforcement hit logs; simulation vs production diff
03.01.03	Information flow enforcement for CUI (CUI flow between connected systems)	Boundary policies CUI ↔ non-CUI; ADM-approved paths	ADM workspace export; boundary rule set; denied-flow report for violations
03.01.05	Least privilege (incl. network)	Minimal allow-list; process-scoped visibility	Rule review with owner/ticket metadata; process-attributed flows
03.01.12	Remote access	Restrict remote paths to managed gateways; log allowed admin sessions	Policies targeting jump hosts / bastions; flow logs for admin ports
03.03.01	Event logging	Flow + process telemetry	SIEM feed samples; retention configuration record
03.03.02	Audit events	Detailed connection records	Exported flows with 5-tuple, process, user context
03.03.03	Audit record content	Rich telemetry	Field mapping doc (see Appendix B)
03.03.04	Response to audit logging process failures	Alert when sensor/connectors stop; SIEM ingest health dashboards	SOC reference designs; tickets for logging pipeline failures

Requirement	Title / intent (summary)	CSW capability	Evidence produced
03.03.05	Audit record review, analysis, and reporting	Export to SIEM; correlation with identity and netflow	SOC review cadence; sample investigation reports
03.03.06	Audit record reduction and report generation	Flow/process summaries for investigations	Saved queries / reports used for after-action review
03.03.07	Time stamps (UTC granularity)	Telemetry timestamps aligned to enterprise time standard	Sample records + NTP configuration reference
03.03.08	Protection of audit information	RBAC on CSW console; least-privilege admin to change logging	Admin access list; change control for log integrations
03.04.01	Baseline configuration	ADM as communication baseline	Baseline vs current diff after change window
03.04.02	Configuration change	New flows / listeners flagged	Change tickets linked to observed deltas
03.04.08	Authorized software — allow by exception	Process / software inventory; block unexpected listeners via policy	Approved software list vs CSW inventory reconciliation
03.11.x	Risk assessment inputs	CVE + reachability + EPSS-style prioritization if enabled	Risk workshop slides + CSW export attachments
03.13.01	Boundary protection	Enclave segmentation; internal key interfaces controlled	Topology from ADM; internal boundary rule list
03.13.06	Deny by default / allow by exception	Workload-level default deny	Signed allow-list; percentage of traffic denied as “unknown” near

Requirement	Title / intent (summary)	CSW capability	Evidence produced
03.13.08	Transmission and storage confidentiality	Detect/block prohibited cleartext paths; visibility into sensitive flows (transmission) ; <i>encryption at rest</i> is validated outside CSW	zero post-tuning Protocol analysis exports; blocked-flow metrics; SSP narrative for TLS/IPsec coverage
03.14.01	Flaw remediation (visibility / prioritization support)	Vulnerability exposure reports; reachability-aware prioritization	POA&M extracts; weekly export samples
03.14.02	Malicious code protection (complementary signal)	Process lineage; anomalous egress in addition to AV/EDR	CSW process-tree exports correlated with AV findings
03.14.06	System monitoring	Workload network + process monitoring; unauthorized connection detection	Alert catalog; dashboard screenshots; SOC coverage statement
03.06.x	Incident handling (technical evidence)	Scope and timeline reconstruction	Forensic export bundle per IR playbook
03.12.03	Continuous monitoring	Ongoing workload telemetry + policy effectiveness metrics	Monitoring strategy attachment referencing CSW dashboards
03.12.05	Information exchange agreements (technical interfaces)	ADM shows live interfaces; policy enforces approved interconnections only	Interface list from ADM + signed exchange agreement references in SSP only

11. Common Pitfalls

Pitfall	Mitigation
CUI workload missing from CSW inventory	Weekly reconcile against SSP; auto-label from cloud metadata where possible
Over-broad rules (subnet-only) undermine 03.01.05 evidence	Prefer workload + process context from ADM; document exceptions
ADM window too short (misses batch jobs)	Run ≥ 14 days; include month-end and patch cycles
Silent cleartext on non-standard ports	Inspect process-level protocol data; do not assume encryption by port number alone
Enforcement without successful simulation	Complete simulation with CAB sign-off; stage by enclave
Logs retained only in CSW console	Forward to SIEM / immutable store for assessor retention tests

12. Boundaries — What CSW Does **Not** Cover

- **Policy, training, personnel, and HR workflows** (families 03.02, 03.09, parts of 03.15–03.17 documentation obligations).
- **Physical security** (03.10) and **media handling** procedures (03.08).
- **Cryptographic module validation** (FIPS 140) — CSW may help you see **where** crypto should be enforced, not **certify** implementations.
- **Authoritative identity lifecycle** (account provisioning, MFA enrollment, PIV issuance) — CSW consumes identity context where integrated but does not replace IdP/IAM.
- **Formal security assessments** — CSW supplies artifacts; assessors render official findings.
- **Out-of-scope workloads** — if an asset is missing an agent or connector, CSW cannot assert controls for it.

13. Audit Preparation and Evidence Export

13.1 Quarterly Evidence Bundle (Suggested Contents)

Artifact	CSW source	Notes
Scope membership export	Inventory	Must reconcile to SSP component list
ADM baseline snapshot	ADM workspace	Time-bounded, signed by owner
Active policy workspace	Defend	Version ID + approver

Artifact	CSW source	Notes
Enforcement / simulation report	Policy analytics	Show drift vs prior quarter
Vulnerability exposure report	Vulnerability	CUI scope only
Sample SIEM ingest	SOC	Proof of central retention

13.2 Optional CLI / Automation Examples

Use your enterprise automation pattern (CI pipeline, SOC playbook). Example: **periodic inventory pull** via your supported API/automation tool (replace with your gateway host and tokens):

```
#!/usr/bin/env bash
set -euo pipefail
# Illustrative only – use the official Cisco Secure Workload API /
# automation documented for your tenant.
export CSW_API_BASE="https://<tenant-api-endpoint>"
export CSW_API_TOKEN="$(vault kv get -field=token
secret/csw/automation)"
curl -sS -H "Authorization: Bearer ${CSW_API_TOKEN}" \
    "${CSW_API_BASE}/v2/inventory/export?scope=CUI-Enclaves" \
    -o "cui_inventory_$(date -u +%Y%m%dT%H%M%S).csv"
```

Store exports in protectively marked storage consistent with CUI handling rules.

Appendix A — Cross-Walk to NIST SP 800-53 Rev. 5 (Illustrative)

800-171 Rev. 3	Related 800-53 control(s) (per NIST mapping)
03.01.03	AC-4
03.01.05	AC-6
03.13.01	SC-7
03.13.06	SC-7(5)
03.03.x	AU-2, AU-3, AU-6, AU-12 (family-dependent)
03.14.x	SI-4, SI-7 (representative)

Use official NIST mapping tables in **SP 800-171** and **SP 800-171A** as the authoritative cross-walk.

Appendix B — SIEM Field Mapping (Example)

Map exported events to your logging standard. Minimum useful fields:

- Timestamp (UTC), sensor/workload ID, source/destination IP and port, protocol, action (allow/deny/alert), policy ID, process name, process hash, user (if available), scope labels (cui_scope, enclave_id).
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Appendix C — Related Framework Reference designs

- [NIST SP 800-53 Rev. 5](#) — federal control catalog alignment at AC, AU, CM, IR, RA, SC, SI.
 - [CMMC 2.0](#) — assessment-oriented CUI scope patterns and Level 2 practice mapping (verify Rev. 2 vs. Rev. 3 contract baseline).
 - [NIST SP 800-207](#) — Zero Trust architectural context for segmentation and telemetry.
 - [FedRAMP](#) — when CUI processing shares infrastructure with FedRAMP-style packages.
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Replace organization-specific names and bracketed placeholders before external distribution. For contractual compliance, rely on your legal and program office interpretation of applicable NIST revisions.